

1.Case Study: Monthly Hospital Patient Visits

A hospital administrator wants to analyze the number of patients visiting the hospital each month during 2025.

Month	Patient Visits
Jan	1200
Feb	1350
Mar	1500
Apr	1450
May	1700
Jun	1850

Questions

1. Which visualization is most suitable?
2. What trend can be observed?
3. Identify the month with the highest patient count.
4. Predict whether visits are increasing or decreasing.

2.Case Study: Study Hours vs Exam Marks

A teacher wants to determine whether studying more hours improves exam performance.

Student	Study Hours	Marks
A	2	45
B	3	52
C	4	58
D	5	68
E	7	82

Questions

1. Which chart is appropriate?
2. Is there a positive or negative relationship?
3. Which student is an outlier, if any?
4. Can marks be predicted from study hours?

3. Case Study: Employee Salary Distribution

A company wants to understand how salaries are distributed among employees.

Salary Data (₹ in Thousands)

25, 28, 30, 32, 35, 35, 36, 38, 40, 42, 45, 48, 50, 52, 55, 60, 65, 70

Questions

1. Which graph should be used?
2. In which salary range do most employees fall?
3. Is the distribution symmetric or skewed?
4. Are there any salary ranges with very few employees?

4. Case Study: Favorite Programming Language

A survey was conducted among 200 students.

Language	Number of Students
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Python	80
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Java	50
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C++	30
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JavaScript	25
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R	15
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Questions

1. Which visualization is suitable?

2. Which language is most popular?
3. Which language is least preferred?
4. Compare Python and Java popularity.

5. Case Study: Household Monthly Expenses

A family spends money in the following categories:

Category	Expense (%)
Rent	40
Food	25
Education	15
Transportation	10
Entertainment	10

Questions

1. Which chart best represents the data?
2. Which category consumes the largest budget?
3. What percentage is spent on Food and Education together?
4. Which categories occupy the smallest share?

6. Case Study: Student Test Scores

Scores obtained by students:

45, 50, 52, 55, 58, 60, 62, 65, 68, 70, 72, 75, 78, 80, 95

Questions

1. Which plot should be used?
2. What is the median score?
3. Are there any outliers?

4. What is the spread of the data?